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Voltage Estimation of Active Distribution Network Using PMU Technology

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ABSTRACT As the Distributed Generation (DG) is evolving, the distribution network has undergone tremendous change. The DG sources like wind and solar are intermittent in nature. Hence to estimate the Active distribution network's (ADN) parameters (Voltage and Current) accurately with the DG variation, the traditional offline method like long-term smart distribution network planning (with off-line variant) method is not feasible in terms of network's reliability and operation. In this work, an online voltage estimation technique using PMU (Phasor Measurement Unit) technology is proposed to enhance a distribution network's reliable operation and real-time monitoring. The proposed estimation method does not require PMU units at all nodes in the network. The estimation of unknown states information can be achieved from available PMU unit's data. The proposed methodology is economical and feasible for voltage estimation, and system observability as the number of PMU units required is less. The standard IEEE Distribution network systems are considered for checking the feasibility of the recommended technique. For the simulation of the case studies, MATLAB programming is used. A Forward and Backward sweep (FBS) load-flow algorithm is used to authenticate the proposed methodology.

INDEX TERMS Distributed power generation, optimization, phasor measurement unit.

I. INTRODUCTION

With the rise of distributed or embedded generation in Distribution networks (DN) and local energy producers' involvement, distribution systems have become active and grown more complex. The deregulation market introduced competition among new players, bringing a two-way flow of power in distribution systems. These happenings made the distribution network transform from passive to active configuration; in turn, their difficulty and changing aspects have been built up over time [1]. This scenario demands advanced management systems with accurate monitoring, control, and adequate protection systems for reliable operation, which is a most significant task for the distribution network operators [2]. Distribution networks typically have quite a lot of buses. Nevertheless, it is not practical to furnish all the distribution network nodes with metering equipment from an economic perspective. In this context, DSE (Distribution state estimation) plays a predictable part in the actual measurement and monitoring of distribution networks [3]. Typically, due

to the high R/X value of the distribution network, high DG penetration, lack of proper communication exchange among the nodes, and other factors, the active distribution network's state estimation process is a strenuous task compared to the transmission networks [4].

Various works that have been contributed to the literature to overcome the above issues are discussed as follows. For updating the change of events in the network states of a distribution system, the micro PMU's recorded data is being used in [5]. Multi-area parameter estimation is proposed as a solution for distribution grid monitoring issues in [6]; this methodology depends on an improved version of the weighted least squares (WLS) method, which permits to know how the information is shared among the different overlapping nodes with different neighboring zones. An online method to estimate the parameters using PMU measurements based on the least square estimation technique has been proposed in [7]. This paper contributes an online estimation technique to measure voltage and current states under high DG penetration [8]. Reference [9] utilizes error covariance in distribution system state estimation to provide a technique for obtaining optimal measurement locations.

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A gradient-based multi-area algorithm is proposed for the distribution network state estimation, and it is expressed through a weighted least squares problem in [10]. The authors of [11], proposed a methodology to estimate the target bus voltage with known bus information using a supervised learning algorithm. A neural network technique that offers robustness for any restructuring in the network topology discussed in [12]. A numerical method has been proposed in [13] to classify the distribution system's topology and estimate line parameters. Reference [14] explained a technique for estimating the parameter for unbalanced and un-transposed distribution lines. This work addresses the state estimation problem in an unbalanced distribution system by exploiting virtual reference buses [15]. Singh *et al.* [16] recommended a Monte Carlo simulation-based probabilistic technique for placement of meters to reduce the relative errors of estimated voltage and angle of the buses in the network. To identify the best location for meter placement, a site with a significant 2- error ellipse area is considered. An ordinal optimization has been proposed, which gives precise calculations of the possibilities associated with it by decreasing the highness of the search space. As the valuation of the exact objective function is computationally exclusive, they conducted a primary quest to evaluate the approximate objective function; from this, the best results are chosen for the exact objective function [17]. The authors in [18] suggest a stochastic approach for finding the best feasible meter location for state estimation of the distribution network. This method compares the estimation quality with and without placing the meter to check whether a particular site suits the meter location. In [19], the authors proposed a Dynamic programming-based optimization technique for optimal selection and placement of measurement devices. As the uncertainty in the available information risks the operating conditions of the network, they suggested the Monte Carlo method to take into account uncertainties in the load demand, network parameters, and measuring devices. Reference [20] suggested a Gaussian component combination approach to integrate non-monitored DG generation and estimate the optimal meter placement for the distribution state estimation. The non-monitored DG generation is characterized by non-Gaussian probability density distribution and estimated by the Gaussian Mixture Model. In [21], the authors used the Fisher Information Matrix method to estimate the unknown states of the system from the available measurement unit data for the best meter placement.

The above literature has contributed valued research work which deals with state estimation of distribution network. The concept of zero injection measurements, in addition to the PMUs, is considered widely for PMU placement problem constraint formulation for the analysis of network observability. Though the zero injections minimize the required number of PMUs, however, the limitations associated with the zero injection Bus (ZIB- neither generation nor load) concept for PMU placement in ADN have not been well addressed. As the distribution networks are heavily loaded, and the loads are highly unpredictable, and also with the gradual evolvement of

DG into it, none of the nodes/buses can be ignored for accurate estimation of the states of DN in terms of monitoring, study, and enhance the network's operating condition.

This paper aims to propose a methodology for estimating the voltage magnitude of the radial distribution network using PMU technology. An integer linear programming (ILP) algorithm is adopted to define the optimal number and location of PMUs. In addition to the proposed voltage estimation technique, an algorithm is developed to validate the ILP functioning in radial test feeders for complete observability of the system with a minimum number of PMUs.

The significant offerings of this paper are explained as follows. 1) As omission of ZIBs from the optimal placement problem leads to missing some of the dominant nodes in distribution networks, this methodology uses only PMU measurements to analyze voltage estimation without considering zero injections. Hence, the entire observability of the system is obtained without excluding any nodes in the DN. 2) This technique is befitted to any radial DN structure disregarding the feeder length, number of laterals and sub laterals, and variability of X/R ratios. 3) The PMU positions are designated exclusively based upon the network topology. These positions are unaltered by the location of DGs and any other voltage regulating equipment installation. 4) This method uses information from available PMU devices in the network to estimate the voltage magnitude of unmeasured buses, and hence whole network observability is accomplished with the minimal number of PMUs.

The workflow of this paper is described in the following sections as follows. The PMU placement in the distribution network and its validation is discussed in Section II. The proposed voltage profile estimation technique for an active distribution system is explained in section III. In section IV Simulation results of case studies are presented. Conclusion of this paper is given in section V.

II. PMU PLACEMENT IN DISTRIBUTION NETWORK USING INTEGER LINEAR PROGRAMMING (ILP)

A Phasor measurement unit (PMU) is employed to estimate voltage or current magnitude and phase angle in the power system network utilizing a standard time base for synchronization. The resultant measurement is recognized as a synchrophasor. Synchronized measurement sets acquired from synchronized phasors be responsible for an enormously upgraded technique for tracing power system dynamic behavior for enhanced electrical system protection, monitoring, stability, operation, and control [22].

A. PMU PLACEMENT

For siting the PMUs optimally, recognizing the ideal sites is the primary phase. In a practical electrical network framework, certain nodes might be distinct and critical, for instance, a node accompanying an overloaded or reasonable predominant site, or could be an upcoming expansion location. Owing to former occasions, at the preferred sites, PMUs could be situated. The non-observable phasors of the system

could be observable with few extra PMUs. A binary search algorithm is exploited to explain the PMUs sites optimally, which would permit the complete network to be observable under usual operating events [23].

B. OBSERVABILITY ANALYSIS UNDER NORMAL CONDITONS

This work uses a topological method to authenticate the observability of the entire power distribution network [24].

It is acceptable; if the bus type and topology of the network are identified to site the placing PMUs optimally. The incidence matrix represents the bus interconnection of the network. The matrix E exemplifies the correlation in the buses by lines/branches of the system. If E and L constitute the connectivity matrix and total nodes of the network, E_p is a $L \times L$ matrix, and the elements $([e]_{pq})$ of the matrix are explained in equation (1) [25].

$$[e]_{pq} = \begin{cases} 1, & \text{if } p = q \\ 1, & \text{if buses } p \text{ and } q \text{ are linked} \\ 0, & \text{else} \end{cases} \quad (1)$$

The interrelation setting of the PMUs at the p_{th} placement bus is explicated by equation (2).

$$y_p = \begin{cases} 1, & \text{if a PMU is situated at } p_{th} \text{ bus} \\ 0, & \text{else} \end{cases} \quad (2)$$

The objective function of this scenario to achieve complete system observability by optimal placement of PMUs is explained by equation (3) [26].

$$\min \sum_{p=1}^L y_p, \quad p = 1, 2, 3, \dots, L \quad (3)$$

An 11-bus radial distribution feeder is considered to elucidate the observability of the whole system and is demonstrated in Figure 1.

The incidence matrix associated to Figure 1 is indicated in equation (4).

$$E = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad (4)$$

1) PROBLEM CONSTRAINTS OF THE SYSTEM

By presuming, the nodes/buses at which PMUs are situated are assigned with adequate input data to compute the voltage and current phasors of the network. A node in the network is observable; at the minimum, one measuring unit is situated at

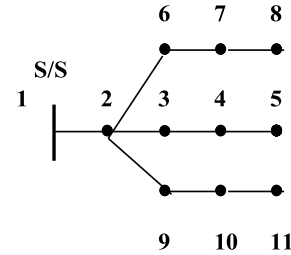


FIGURE 1. 11-Bus radial test feeder topology.

the advised site or at any of its adjacent nodes. Hence, the p_{th} node observability is described by equation (5) [27].

$$h_p = \sum_{q=1}^L e_{pq} y_q \geq 1, \quad p = 1, 2, 3, \dots, L \quad (5)$$

The restructured form of equation (5) in matrix form is given below:

$$H = EY \geq s \quad (6)$$

where,

$$Y = [y_1, y_2, y_3, \dots, y_L]^T_{L \times 1} \quad (7)$$

$$s = [1, 1, \dots, 1]^T_{L \times 1} \quad (8)$$

$$H = [h_1, h_2, \dots, h_L]^T_{L \times 1} \quad (9)$$

The requisite constraints to assure the full observability of the 11-bus feeder from Figure 2, are given by equation (10).

$$\begin{cases} h_1 = y_1 + y_2 \geq 1 \\ h_2 = y_1 + y_2 + y_3 + y_6 + y_9 \geq 1 \\ h_3 = y_2 + y_3 + y_4 \geq 1 \\ h_4 = y_3 + y_4 + y_5 \geq 1 \\ h_5 = y_4 + y_5 \geq 1 \\ h_6 = y_2 + y_6 + y_7 \geq 1 \\ h_7 = y_6 + y_7 + y_8 \geq 1 \\ h_8 = y_7 + y_8 \geq 1 \\ h_9 = y_2 + y_9 + y_{10} \geq 1 \\ h_{10} = y_9 + y_{10} + y_{11} \geq 1 \\ h_{11} = y_{10} + y_{11} \geq 1 \end{cases} \quad (10)$$

From the above expression, a plus (“+”) sign represents arithmetical summation, and one (“1”) signifies that at least one variable in the summation is expected to be non-zero. For instance, the inequality constraint from the expression (7) specifies that at the least one measuring unit intended to be placed at the first or second node (or the two of) for bus one to have complete observability. The solution for optimal PMU placement of Figure 1, using ILP is $Y = [1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0]^T$, ensures the entire system observable by employing PMUs at buses 1, 4, 7, and 10.

ILP is a mathematical optimization technique wherein specific or entire constrained variables are restricted to be either binary or integers. ILP includes mainly three effective formulations; 1) Variables; execute the decisions regarding integer

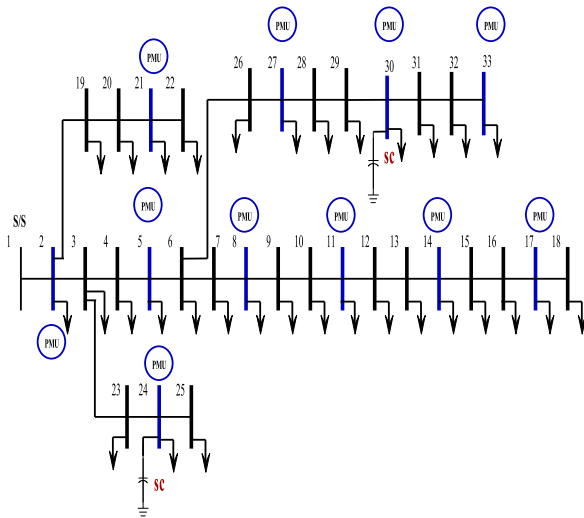


FIGURE 2. 33 Bus system with PMUs, DG and SC integration.

values, 2) constraints; enclose the costs to a manageable region; and 3) the objective function; gets the optimal solution by directing the problem by maximizing or minimizing its function. The objective function and constraints need to be linear consistently in ILP formulation. This technique affirms the accomplishment of complete observability of the network with optimum PMU numbers.

The technique utilized for locating the optimum placement of PMU is analyzed in Algorithm 1.

Algorithm 1 Algorithm of Optimum PMU Location

- 1: Initialize the variables
- 2: Enter the details of load data and total number of buses of the designated network
- 3: Build the incidence matrix (E)
- 4: Input the constraints to set bounds for lower and upper limits
- 5: Execute the intlinprog function to acquire the optimal PMUs number and corresponding locations
- 6: Present the PMU number and positions
- 7: End

C. METHODOLOGY FOR VALIDATION OF OPTIMAL PLACEMENT PROBLEM

Validation was done for the results obtained by creating an algorithm using MATLAB code that checks whether all the buses are connected to the PMUs, which is the whole purpose of an optimal placement solution. All the PMU buses and the remaining buses are sorted into two different arrays and checking all the buses connected to PMU buses through branch array and collecting them in Val array (Val array is initiated as a null array and filled with the bus numbers that are connected to the PMU buses). The system is fully observable if all the remaining bus numbers are there in the Val array; if not, the system is not fully observable, and the

Algorithm 2 Validation of Optimal Placement Problem

- 1: Input the number of buses (n), branches (b), PMU bus locations (p), remaining buses (r)
- 2: Initialize $i = 1, r = 1, j = 1$
- 3: **if** PMU[i] = 1 **then**
- 4: PMUbus [j] = i
- 5: $j = j + 1$
- 6: $i = i + 1$
- 7: **else** rembus [r] = i
- 8: $r = r + 1$
- 9: $i = i + 1$
- 10: **end**
- 11: **if** $i \leq n$ **then**
- 12: **go to** step 2
- 13: **else** initialize $i = 1, a = 1$
- 14: initialize $j = 1$
- 15: Check for each PMU bus element in the branch array's first column and then second column
PMUbus [i] = branch [j, 2] and PMUbus [i] = branch [j]
- 16: If PMU bus elements are found in branch array, include the other elements in the row into val array
val[a] = branch [j, 1] and val[a] = branch [j, 2]
- 17: **end**
- 18: **end**
- 19: Initialize $i = 1, c = 1$
- 20: initialize $j = 1$
- 21: **if** rembus[i] = val [j] **then**
- 22: check [c] = rembus [i]
- 23: $c = c + 1$
- 24: $j = j + 1$
- 25: **else go to** step 24
- 26: **if** $j \leq r$ **then**
- 27: **go to** step 21
- 28: **else** $i = i + 1$
- 29: **end**
- 30: **end**
- 31: **if** $i \leq r$ **then**
- 32: **go to** step 20
- 33: **else if** check = rembus **then**
- 34: Given system is fully Observable
- 35: **else** Given system is not fully Observable
- 36: **end**
- 37: **end**

PMU optimal locations which are identified would be wrong. Step by detailed step methodology of the validation process of optimal placement is explained in Algorithm 2.

III. METHODOLOGY OF PMU MEASUREMENT SETUP FOR VOLTAGE PROFILE ESTIMATION

A generalized code for N number of buses is developed to give the PMU nodes' voltages and the currents in the branches connected to PMUs. End_bus array consists of all the bus numbers of the ones at the end of every lateral chain (chains

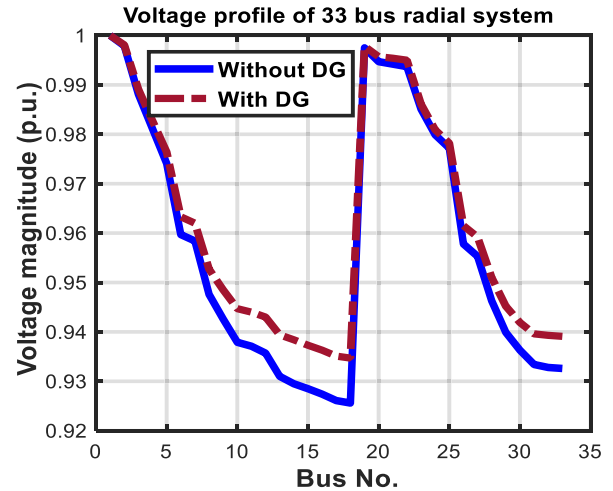
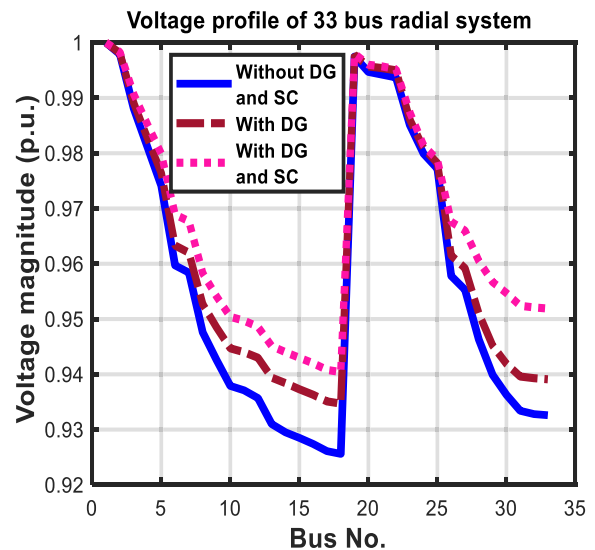
Algorithm 3 PMU Measurement Setup Algorithm

- 1: Start
- 2: Get the PMU locations and branch array.
- 3: Get the active, reactive (P_a , Q_a) powers at each bus and branch impedances (Z).
- 4: Initialize voltage at every bus as $V(j,1) = 1$ p.u.
- 5: Create Endbus array consisting of all the end located buses $\text{Endbus} = \text{zeros}(j,1)$.
- 6: Calculate branch currents using the branch impedance for end branches and then for remaining ones $i(j,1) = \text{conj}(\text{abs}(\text{apparent power})/\text{abs}(v(j,1)))$.
- 7: Calculate new bus voltages using the branch currents $V_{\text{new}} = V(j,1)$.
- 8: Find the branch currents again using the new bus voltages $i_{\text{new}} = [i]$
- 9: Find the branch currents again using the new bus voltages
- 10: Continue the loop until the difference between the voltage sets is very low ($< \text{tolerance}$)
- 11: Currents and voltages at PMUs are found by checking for the connections in the branch
- 12: Check for the corresponding values in bus voltage and branch current array $V(\text{pmubus}(k))$, $i(\text{pmubus}(k)-1)$.
- 13: Display the bus voltage and branch current arrays
- 14: Stop

TABLE 1. Optimal PMU nodes for IEEE radial bus systems.

IEEE Test System	No. of PMUs	PMU located buses	CPU time (Sec)
33 bus	11	2,5,8,11,14,17, 21,24,27,30,33	0.389 279
69 bus	24	2,6,9,14,17,20,23,26,29,32,35, 37,40,43,46,47,50,52,55,58,61, 64,66,68	0.407 024
119 bus	42	3,5,8,11,14,17,20,23,26,29,32, 34,37,39,42,45,48,51,54,56,59, 62,63,67,70,73,76,78,81,84,87, 90,91,94,98,101,104,107,110, 113,115,118	0.435 688

of buses which are not the main chain, but are connected to it) and buses' main chain (chain of buses from the source). Firstly voltage is assumed at every bus as 1 p.u. (Per Unit). The currents are calculated in branches of end buses using the end_bus array and using Kirchhoff's current law for the remaining ones. Then proceeded to find voltages at the buses by going backward, from the end buses to the first bus by using branch currents from before, in Kirchhoff's voltage law and then finding the branch currents again by using the bus voltages from before and the loop goes on. The voltage magnitudes from this set and the next set, and if the difference is extremely low (< 0.0001), this set would be the desired result, and the loop will be stopped. The voltage magnitudes and currents at PMUs are found by checking the branch array with

**FIGURE 3.** 33 Bus radial system's voltage profile with and without DG connection using PMU data.**FIGURE 4.** 33 Bus radial system's voltage profile with/without DG and SC connection using PMU data.

all the bus connections and finding the magnitudes of voltages and currents for the same in the voltage magnitude array and branch currents array, respectively. The methodology of the PMU measurement setup process is explained in Algorithm 3.

IV. RESULTS AND DISCUSSION

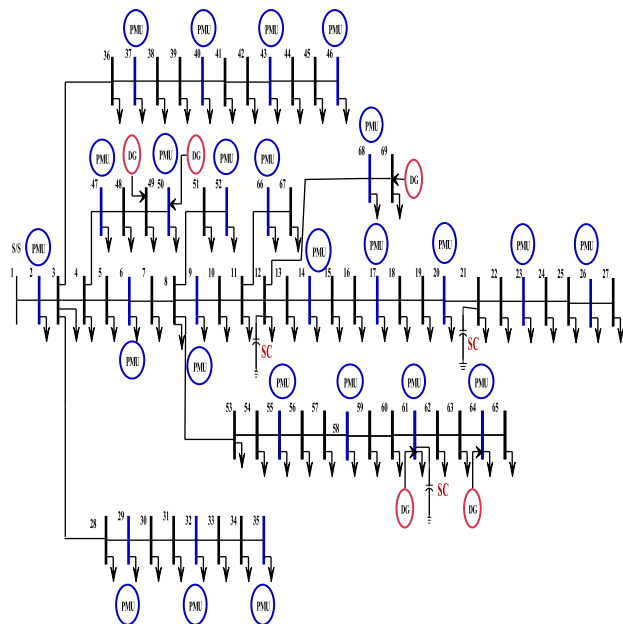
The results of optimal PMU placement and their validation, voltage profile estimation with and without DG using PMUs are explained in this section. The simulations are executed in MATLAB environment on an Intel(R) Core(TM) i5-6200U CPU @ 2.30 GHz, 2400 MHz, 2 Core(s), and 4 Logical Processor(s).

A. OPTIMAL PMU LOCATIONS

The simulation is done using an ILP technique for complete observability of the distribution test feeders with minimal number of PMUs. MATLAB programming is used to perform

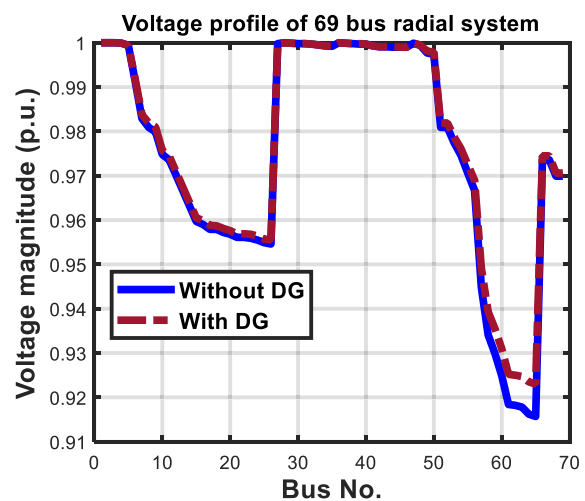
TABLE 2. Validation results of PMU locations radial distribution test systems.

IEEE Test system	Pmu_bus	Rem_bus	Val_array	Check_array
33 bus	[2,5,8,11,14,17,21,24,27,30,33]	[1,3,4,6,7,9,10,12,13,15,16,18,19,20,22,23,25,26,28,29,31,32]	[3,6,9,12,15,18,19,22,25,28,31,1,4,7,10,13,16,20,23,26,29,32]	[1,3,4,6,7,9,10,12,13,15,16,18,19,20,22,23,25,26,28,29,31,32]
69 bus	[2,6,9,14,17,20,23,26,29,32,35,37,40,43,46,47,50,52,55,58,61,64,66,68]	[1,3,4,5,7,8,10,11,12,13,15,16,18,19,21,22,24,25,27,28,30,31,33,34,36,38,39,41,42,44,45,48,49,51,53,54,56,57,59,60,62,63,65,67,69]	[3,7,10,15,18,21,24,27,30,33,38,41,44,48,53,56,59,62,65,67,69,1,5,8,13,16,19,22,25,28,31,34,36,39,42,45,4,49,51,54,57,60,63,11,12]	[1,3,4,5,7,8,10,11,12,13,15,16,18,19,21,22,24,25,27,28,30,31,33,34,36,38,39,41,42,44,45,48,49,51,53,54,56,57,59,60,62,63,65,67,69]
119 bus	[3,5,8,11,14,17,20,23,26,29,32,34,37,39,42,45,48,51,54,56,59,62,63,67,70,73,76,78,81,84,87,90,91,94,98,101,104,107,110,113,115,118]	[1,2,4,6,7,9,10,12,13,15,16,18,19,21,22,24,25,27,28,30,31,33,35,36,38,40,41,43,44,46,47,49,50,52,53,55,57,58,60,61,64,65,66,68,69,71,72,74,75,77,79,80,82,83,85,86,88,89,92,93,95,96,97,100,102,103,105,106,108,109,111,112,114,116,117]	[6,9,12,15,18,21,24,27,30,35,40,43,46,49,52,57,60,64,68,71,74,77,79,82,85,88,91,92,95,96,99,102,105,108,111,112,116,2,4,7,10,13,16,19,22,25,28,31,33,36,38,41,44,47,50,53,55,58,61,1,66,69,72,75,65,80,83,86,89,90,93,97,100,103,106,109,112,114,117]	[1,2,4,6,7,9,10,12,13,15,16,18,19,21,22,24,25,27,28,30,31,33,35,36,38,40,41,43,44,46,47,49,50,52,53,55,57,58,60,61,64,65,66,68,69,71,72,74,75,77,79,80,82,83,85,86,88,89,92,93,95,96,97,99,100,102,103,105,106,108,109,111,112,114,116,117]

**FIGURE 5.** 69 Bus system with PMUs, DG and SC integration.

the case studies on IEEE 33-bus, 69-bus, and modified 119-bus network, respectively. The whole observability of these test feeders with the requisite PMU numbers and their locations are shown in Table 1.

From Table 1, it is noticed that for the network to be fully observable, there is no necessity of locating PMUs at all

**FIGURE 6.** 69 Bus radial system's voltage profile with and without DG connection using PMU data.

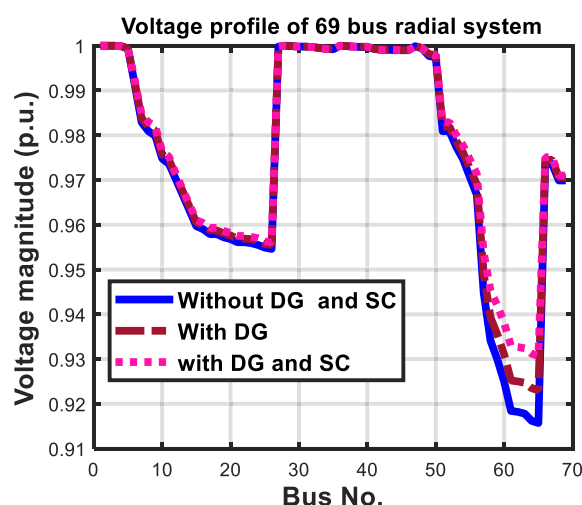
the buses in the system. Henceforth, as the least number of PMUs are needed, it is economical, adequate, and effective in obtaining optimal placement and system observability.

B. VALIDATION RESULTS OF PMU LOCATIONS

The Validation results of PMU Locations of 33, 69 and 119 radial test systems are presented in Table 2. As both the check and rem_bus arrays are the same in all test systems, given systems are completely observable.

TABLE 3. Comparison of 33 bus radial system voltage magnitude results of PMU measurement setup with FBS algorithm.

Bus Location	Voltage Magnitude (p.u.) using PMU setup			Voltage Magnitude (p.u.) using FBS Algorithm		
	Without DG and SC	With DG	With DG and SC	Without DG and SC	With DG	With DG and SC
2	0.9979	0.9981	0.9983	0.9978	0.9980	0.9982
5	0.9741	0.9764	0.9799	0.9742	0.9763	0.9797
8	0.9476	0.9527	0.9583	0.9475	0.9526	0.9580
11	0.9371	0.9440	0.9497	0.9370	0.9438	0.9496
14	0.9295	0.9384	0.9440	0.9296	0.9382	0.9440
17	0.9261	0.9351	0.9408	0.9260	0.9350	0.9406
21	0.9942	0.9954	0.9957	0.9940	0.9952	0.9956
24	0.9799	0.9809	0.9814	0.9798	0.9808	0.9816
27	0.9553	0.9592	0.9660	0.9553	0.9592	0.9658
30	0.9363	0.9419	0.9547	0.9360	0.9415	0.9545
33	0.9326	0.9391	0.9519	0.9325	0.9390	0.9518

**FIGURE 7.** 69 Bus radial system's voltage profile with/without DG and SC connection using PMU data.

C. PMU MEASUREMENT SETUP RESULTS OF VOLTAGE PROFILE ESTIMATION

The PMUs are installed in the distribution network test systems based on the results obtained from the optimal PMU locations, as discussed in section I. It is observed that the results obtained from PMU measurements are close to forward backward sweep method results. The rated line voltage for all test systems is considered 12.66 kV and Base MVA = 100. The system data for 33 bus and 69 bus are given in [28], 119 bus data is given in [29].

1) 33 BUS SYSTEM PMU READINGS WTH/WITHOUT DG AND SHUNT CAPACITOR (SC) INTEGRATION

The 33 bus radial distribution network is shown in Figure 2. A total load of this system is 4369 kVA with PF = 0.85. The network has four DGs placed at buses 14, 17, 21, and 32 with a total DG capacity of 371.5 kW and 230 kVAR and two SCs are placed at buses 24 and 30 with a rating of 600 kVAR and 750 kVAR, respectively.

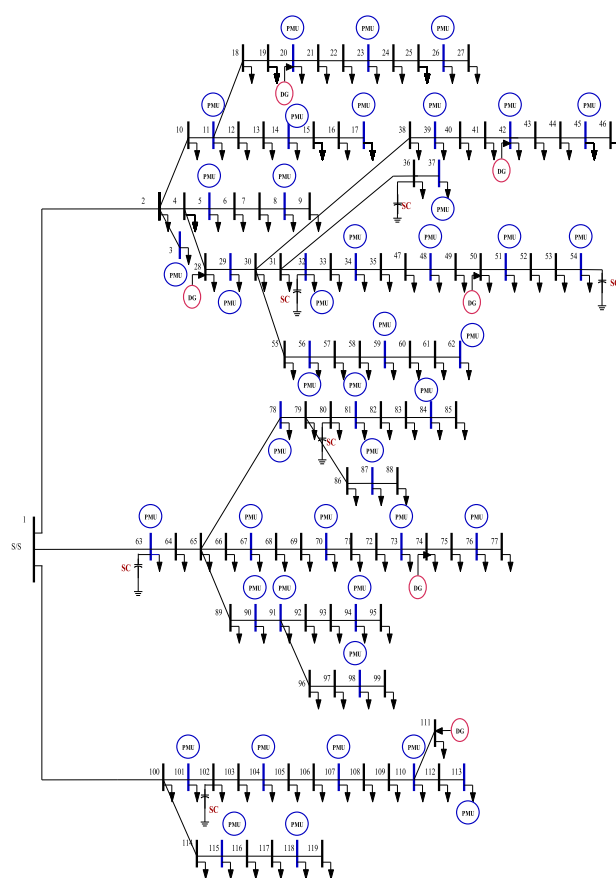
**FIGURE 8.** 119 Bus system with PMUs, DG and SC integration.

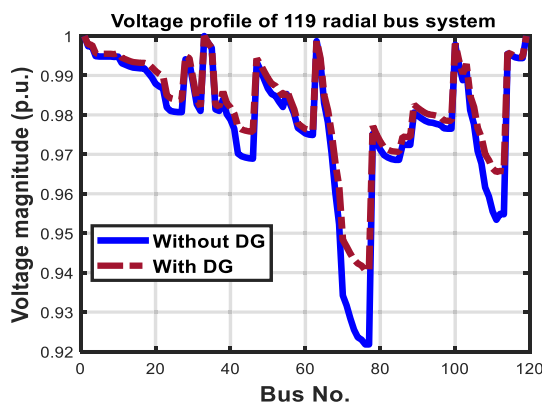
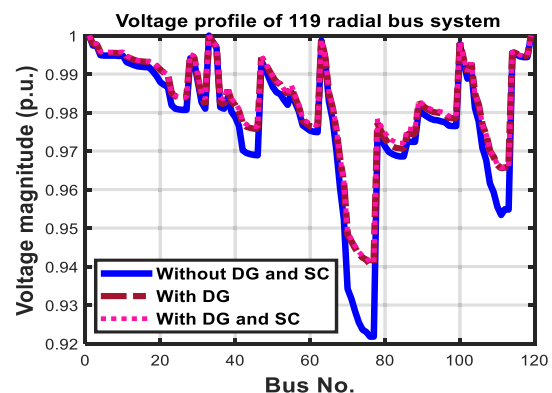
Table 3 gives the comparative voltage magnitude results of 33 bus radial distribution network using PMU setup and FBS load flow algorithm. It is observed that the PMU set up results are matching with the FBS load flow algorithm.

Figure 3, displays the voltage profile of the 33 bus radial distribution system using PMU data. With the DG installation, the voltages at all nodes are improved to an extent.

The voltage estimation analysis is done further by installing SCs along with the DGs in the network.

TABLE 4. Comparison of 69 bus radial system voltage magnitude results of PMU measurement setup with FBS algorithm.

Bus Location	Voltage Magnitude (p.u.) using PMU setup			Voltage Magnitude (p.u.) using FBS Algorithm		
	Without DG and SC	With DG	With DG and SC	Without DG and SC	With DG	With DG and SC
2	1	1	1	1	1	1
6	0.9913	0.9918	0.9922	0.9912	0.9917	0.9920
9	0.9799	0.9810	0.9820	0.9798	0.9808	0.9818
14	0.9632	0.9640	0.9645	0.9634	0.9638	0.9644
17	0.9579	0.9588	0.9593	0.9578	0.9589	0.9591
20	0.9568	0.9576	0.9582	0.9565	0.9575	0.9580
23	0.9559	0.9568	0.9573	0.9557	0.9566	0.9572
26	0.9546	0.9555	0.9560	0.9547	0.9553	0.9561
29	0.9999	0.9999	0.9999	0.9998	0.9998	0.9997
32	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997
35	0.9992	0.9992	0.9992	0.9990	0.9990	0.9990
37	0.9999	0.9999	0.9999	0.9997	0.9997	0.9997
40	0.9997	0.9997	0.9998	0.9995	0.9995	0.9997
43	0.9991	0.9991	0.9991	0.9990	0.9990	0.9990
46	0.9990	0.9990	0.9990	0.9989	0.9989	0.9989
47	0.9999	0.9999	0.9999	0.9998	0.9998	0.9999
50	0.9974	0.9979	0.9980	0.9972	0.9975	0.9979
52	0.9809	0.9819	0.9829	0.9805	0.9820	0.9827
55	0.9706	0.9725	0.9745	0.9706	0.9725	0.9743
58	0.9341	0.9393	0.9453	0.9340	0.9395	0.9451
61	0.9184	0.9252	0.9329	0.9185	0.9255	0.9327
64	0.9162	0.9235	0.9312	0.9160	0.9233	0.9311
66	0.9736	0.9745	0.9751	0.9735	0.9742	0.9750
68	0.9699	0.9706	0.9710	0.9695	0.9702	0.9709

**FIGURE 9.** 119 Bus radial system's voltage profile with and without DG connection using PMU data.**FIGURE 10.** 119 Bus radial system's voltage profile with/without DG and SC connection using PMU data.

Figures 4, 7, and 10 show a remarkable enhancement in the voltage profile with the SCs installation and DGs in ADN, and the losses are reduced by a significant amount, which improved the overall voltage profile of the network.

2) 69 BUS SYSTEM PMU READINGS WTH/WITHOUT DG AND SHUNT CAPACITOR (SC) INTEGRATION

The 69 bus radial distribution system is shown in Figure 5. The total load of this system is 2223 kVA with $PF = 0.89$. The network has five DGs placed at buses 49, 50, 61, 64, and 69 with a total DG capacity of 400 kW and 230 kVAR, and three SCs are placed at buses 12, 21, and 61 with a rating of 300 kVAR, 150 kVAR, and 450 kVAR, respectively.

Table 4, gives the comparative voltage magnitude results of 69 bus radial distribution network using PMU setup and FBS load flow algorithm. It is observed that the PMU set up results are similar to the FBS load flow algorithm.

Figure 6, displays the PMU data voltage profile of the 69 bus radial distribution system with and without DG. With the DG integration into the network, all the node voltages are boosted to a higher value.

3) 119 BUS SYSTEM PMU READINGS WTH/WITHOUT DG AND SHUNT CAPACITOR (SC) INTEGRATION

The 119 bus radial distribution network is shown in Figure 8. The total load of this system is 28,392 kVA with $PF = 0.79$.

TABLE 5. Comparison of 119 bus radial system voltage magnitude results of PMU measurement setup with FBS algorithm.

Bus Location	Voltage Magnitude (p.u.) using PMU setup			Voltage Magnitude (p.u.) using FBS Algorithm		
	Without DG and SC	With DG	With DG and SC	Without DG and SC	With DG	With DG and SC
3	0.9974	0.9978	0.9979	0.9972	0.9975	0.9978
5	0.9948	0.9955	0.9958	0.9945	0.9954	0.9957
8	0.9948	0.9955	0.9957	0.9947	0.9953	0.9956
11	0.9931	0.9945	0.9946	0.9930	0.9945	0.9945
14	0.9920	0.9934	0.9935	0.9918	0.9932	0.9933
17	0.9918	0.9932	0.9933	0.9916	0.9932	0.9933
20	0.9879	0.9911	0.9912	0.9875	0.9906	0.9910
23	0.9819	0.9850	0.9851	0.9817	0.9848	0.9850
26	0.9807	0.9839	0.9840	0.9805	0.9837	0.9838
29	0.9935	0.9944	0.9947	0.9932	0.9942	0.9944
32	0.9810	0.9821	0.9835	0.9809	0.9822	0.9833
34	0.9983	0.9983	0.9984	0.9984	0.9984	0.9984
37	0.9810	0.9821	0.9834	0.9808	0.9819	0.9832
39	0.9802	0.9830	0.9836	0.9800	0.9828	0.9834
42	0.9702	0.9770	0.9776	0.9703	0.9771	0.9774
45	0.9690	0.9758	0.9764	0.9692	0.9756	0.9763
48	0.9914	0.9928	0.9930	0.9912	0.9925	0.9928
51	0.9853	0.9879	0.9881	0.9855	0.9880	0.9880
54	0.9820	0.9847	0.9847	0.9817	0.9845	0.9846
56	0.9826	0.9838	0.9844	0.9825	0.9836	0.9840
59	0.9760	0.9771	0.9777	0.9762	0.9770	0.9775
62	0.9749	0.9761	0.9767	0.9747	0.9760	0.9765
63	0.9987	0.9988	0.9989	0.9985	0.9987	0.9988
67	0.9703	0.9753	0.9758	0.9700	0.9752	0.9755
70	0.9342	0.9483	0.9488	0.9340	0.9482	0.9485
73	0.9255	0.9433	0.9438	0.9253	0.9430	0.9435
76	0.9218	0.9406	0.9411	0.9219	0.9407	0.9409
78	0.9752	0.9772	0.9783	0.9751	0.9770	0.9781
81	0.9698	0.9718	0.9733	0.9696	0.9715	0.9730
84	0.9686	0.9707	0.9721	0.9684	0.9704	0.9720
87	0.9724	0.9744	0.9757	0.9722	0.9743	0.9755
90	0.9798	0.9818	0.9823	0.9796	0.9817	0.9822
91	0.9791	0.9811	0.9816	0.9789	0.9809	0.9814
94	0.9780	0.9800	0.9805	0.9775	0.9797	0.9804
98	0.9765	0.9785	0.9790	0.9764	0.9786	0.9788
101	0.9928	0.9938	0.9940	0.9925	0.9935	0.9938
104	0.9812	0.9848	0.9849	0.9810	0.9845	0.9847
107	0.9677	0.9743	0.9744	0.9675	0.9742	0.9744
110	0.9555	0.9664	0.9665	0.9552	0.9666	0.9663
113	0.9548	0.9657	0.9658	0.9548	0.9655	0.9656
115	0.9951	0.9954	0.9955	0.9952	0.9957	0.9954
118	0.9944	0.9947	0.9948	0.9940	0.9945	0.9947

The network has six DGs placed at buses 20, 28, 42, 50, 74, and 111, with a total DG capacity of 3406 kW and 2556 kVAR, and six SCs are placed at buses 32, 36, 54, 63, 80, and 102 with a rating of 450 kVAR, 450 kVAR, 600 kVAR, 450 kVAR, 450 kVAR, and 300 kVAR respectively.

Figure 9, shows the 119 bus radial distribution system's voltage profile measured from PMU data with and without DG. With the incorporation of DG, all node voltages are improved to an extent.

Table 5, compares the voltage magnitude results of the 119 bus radial distribution network using PMU setup and FBS load flow algorithm. It is witnessed that the PMU setup results are almost identical with the FBS load flow algorithm.

The voltages of PMU node connected buses are measured directly by PMU setup. As the network's line impedances

are known, the remaining node voltages may be estimated through Kirchhoff's current law (KCL), ohm's law and the available PMU results. Hence the cost and maintenance of PMU installation can be reduced to a great extent as it is not essential to connect PMUs at each node in the network.

Table 6, gives the total loss reduction of active and reactive powers of 33, 69, and 119 bus radial networks with/without DG and SC integration. The table shows a reduction of 35.79 kW and 25.63 kVAR with DG and 78.28 kW and 54.26 kVAR with DG and SC in 33 bus system, similarly a reduction of 32.3 kW and 15.3 KVAR with DG and 63.89 kW and 29.15 kVAR with DG and SC in 69 bus system, and 238.48 kW and 150.89 kVAR with DG and 267.42 kW and 186.38 kVAR reduction in 119 bus

TABLE 6. Comparison of load flow results of various IEEE radial distribution networks using PMU data.

	33 bus radial system			69 bus radial system			119 bus radial system		
	Without DG and SC	With DG	With DG and SC	Without DG and SC	With DG	With DG and SC	Without DG and SC	With DG	With DG and SC
Total P_{loss} (kW)	195.54	159.75	117.26	227.96	195.66	164.07	766.55	528.07	499.13
Total Q_{loss} (kVAR)	136.98	111.35	82.72	103.19	87.89	74.04	585.03	431.11	398.65
No. of Iteration count	4	4	4	4	4	4	4	4	4
Computation time (Sec)	0.1448	0.1838	0.2087	0.7329	0.8225	0.6453	0.2729	0.6508	0.2544

TABLE 7. Percentage reduction of power loss of radial distribution systems with DG integration using PMU data.

	Percentage reduction in active and reactive power losses after DG and SC integration					
	33 bus		69 bus		119 bus	
Power loss reduction (%)	With DG	With DG and SC	With DG	With DG and SC	With DG	With DG and SC
Total P_{loss}	18.301	40.03	14.16	28.02	31.11	34.88
Total Q_{loss}	18.70	39.61	14.83	28.24	26.30	31.85

system. The percentage of total power loss reduction is given in Table 7.

V. CONCLUSION

In this work, the proposed methodology is effective in the state estimation of any Active radial distribution network. This technique gives real-time information of distribution networks with DG integration and allied problems in monitoring the system states, unlike the traditional distribution network planning method. PMU is real-time monitoring equipment that evaluates the voltage phasor of the node installed on and the neighboring line current phasors attached to it. This capability of PMU, in turn, makes all its direct neighboring buses observable. Various distribution network test systems are analysed for a steady-state voltage estimation using the proposed PMU based online estimation technique. An Integer Linear Programming approach has been discussed on standard radial distribution systems for optimal PMU allocation. Two methodologies are suggested in this work: to validate the complete observability of the distribution network with the optimal number of PMUs, and another one is to estimate the voltage profile of any radial distribution system. The performance of the suggested estimation methodology is validated with the FBS load flow algorithm on distribution test systems with and without DG and SC interconnection, and it is observed that the results obtained from PMU based measurements are close to the former method results.

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